GalvaPulse™

GP-5000

Instruction and Maintenance Manual

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GERMANN INSTRUMENTS A/S

Emdrupvej 102 — DK-2400 — Copenhagen NV — Denmark

Phone : (+45) 39 67 71 17 — Fax : (+45) 39 67 31 67

E-mail : germann-eu@germann.org - Internet : www.germanninstruments.com

Test Smart - Build Right

1. The GP-5000 Kit

The kit comes in one attaché case containing:

1.1. Reinforcement locator
1.2. Reinforcement conductivity meter
1.3. Measuring black cable, banana-banana
1.4. PSION computer with software and attached electronic box containing the pulse generator. The PSION has a Flash card with a backup of the software
1.5. Handheld Ag/AgCl electrode with 3 meter of cable
1.6. Sponges for electrode (one circular and one bar-sized)
1.7. Bottle with saturated KCl solution
1.8. Two adjustable clamping pliers with banana plug
1.9. Sponge for grinding the electrode rings
1.10. Spare batteries for PSION computer, 4 x 1.5V-AA and 1 x 3V-CR1620.
1.11. Drill bits, 12 mm and 18 mm
1.12. Measuring tape
1.13. Calibration box
1.14. Manual
1.15. GalvaPulse Data Viewing and Reporting Program
1.16. Power Point Presentation
1.17. Emery cloth for grinding of electrode rings
1.18. Two reinforcement adapters
1.19. Allen key, 10 mm
1.20. Hammer
1.21. Chalk
1.22. Cable for data transfer to PC and USB to Serial converter cable.

Delivered separately:

1.23. Cable drum with 15 meters of cable
1.24. Concrete sample with a corroding reinforcement and a stainless steel bar cast-in
1.25. Spray bottle

Extra needed on-site:

1.26. Electricity and a cable and drill hammer

2. Introduction

The galvanostatic pulse measurement technique is a fast polarization technique for determination of the actual corrosion rate at the time of testing. The technique was originally developed because Half-Cell Potential (HCP) measurements at wet and semi-wet constructions were difficult to evaluate due to lack of oxygen. When the Galvanostatic pulse measurements are performed, the HCP, the Corrosion Rate and the Resistance are determined, all in one operation.

As default the equipment apply a current of 25 μA for 5 seconds. These values will in passive areas normally give a reasonable polarization of the reinforcement evaluated by the regularity of the polarization graph. Otherwise, the current has to be increased.

At active areas the current is recommended to be 80-100 μA, again for 5-10 seconds. And at very active areas currents up to 400 μA might be necessary to obtain a reasonable polarization. However always try to use firstly as low current and time as possible to minimize the polarization of the reinforcement and increase them if there is not good response.

The potential graph measured has to exhibit a general shape as shown on figure 1 below, in principle, less steep at the beginning in passive corrosion areas and steeper in active areas.

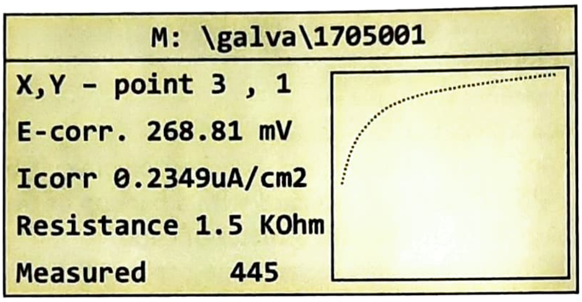


Fig. 1. Galvanostatic pulse response

The essential issue is that the graph, consisting of points, is increasing over time without a large scatter. A decreasing graph indicates a false reading, e.g. caused by a lack of electrical contact, and will not be accepted by the equipment.

A large scatter may indicate bad connections and/or insufficient current pulse and/or time of pulsing.

At passive areas the polarization will change the original half-cell potential in the positive direction for some time. The half-cell potential recorded will be prior to applying the pulse.

At active areas the original half-cell potential will maintain more or less the same value, even after several galvanostatic pulses.

3. Operation instruction of the PSION computer

3.1. Operation of the FORCE program

a. Turn the PSION On by pressing the **On/Esc** key. Fig. 2.

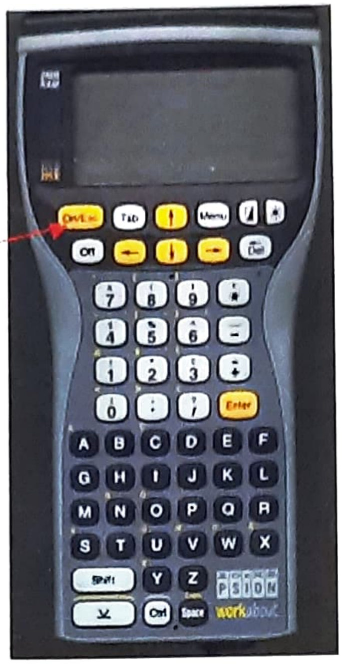


Fig. 2. The PSION computer

b. Select the icon named **FORCE** by using the yellow arrow keys and press Enter.

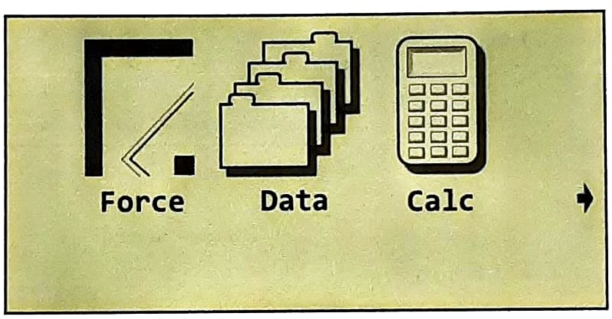


Fig. 3. Display of the FORCE icon

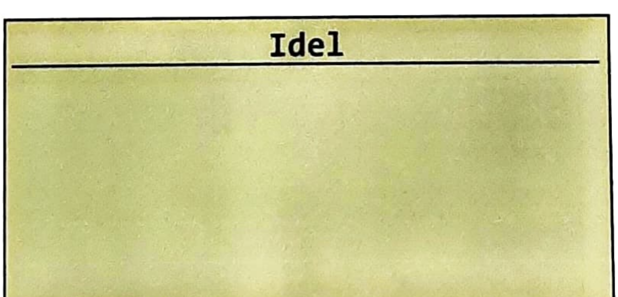


Fig. 4. Display when the application is ready

c. Press the **Menu** button

d. To set the parameters of the instrument prior to a new set of readings select the **Settings** menu. The parameters will be saved and act as default parameters until they are changed again. Fig. 5.

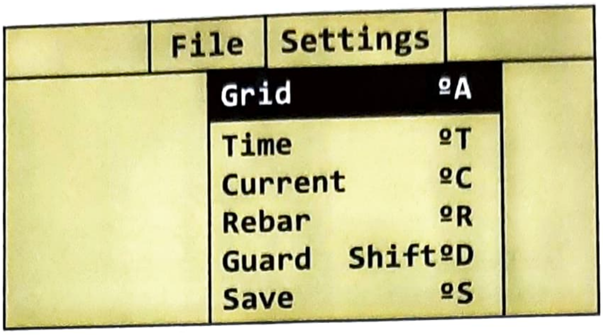


Fig. 5. Settings menu

3.2 Selection of number and spacing of measurement points

a. In the Settings menu the general settings for the measuring area can be changed. Fig. 6 shows the orientation of the X and Y parameters.

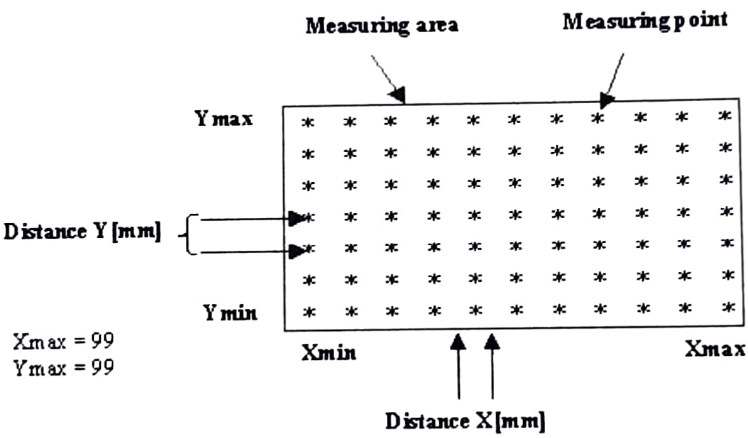


Fig. 6. Sketch of X and Y parameters.

b. Select **Grid** and press **Enter**. Fig. 7.

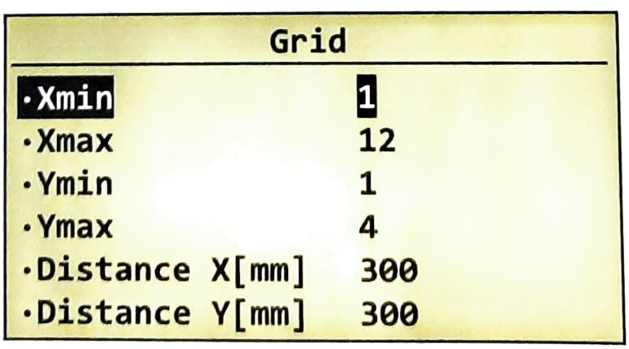


Fig. 7. Grid display

c. Use the arrow keys to select the starting point in the X-direction, **X_{min}**, and the last point in the X-direction, **X_{max}**, (50 is default). Similarly set **Y_{min}** and **Y_{max}** (50 is default).

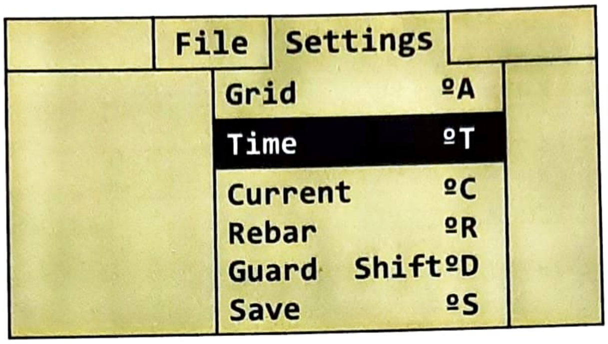
d. If the total number of test points is not known, use the default parameters. Only the grid points with data will be used later in the software.

e. Set the distance between the measuring points in the X-direction, **Distance X** [mm] and in the Y-direction, **Distance Y** [mm] (300 mm is default for both directions).

f. Press **Enter** when the parameters are set.

3.3 Setting pulse duration time.

a. Press the **Menu** key. Fig. 8.



7509-17 HH Crackscope.pdf

Fig 8. Settings display, Time

b. Select **Time** and press **Enter**. Fig. 9.

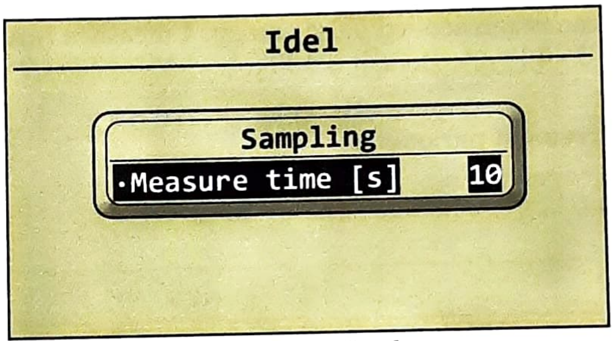


Fig .9 Time display

c. Enter the galvanostatic pulse time in seconds, **Measure time[s],** and press **Enter**. The default value is 5 seconds. Maximum is 20 seconds.

3.4 Setting galvanostatic pulse current

a. Press the **Menu** key to enter the **Settings** menu. Fig. 10.

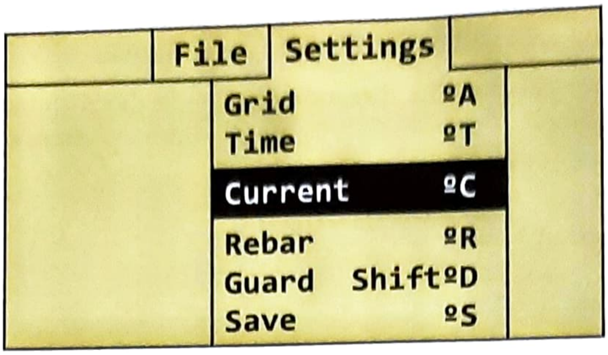


Fig .10. Settings display, current

b. Select **Current** and press **Enter**. Fig. 11.

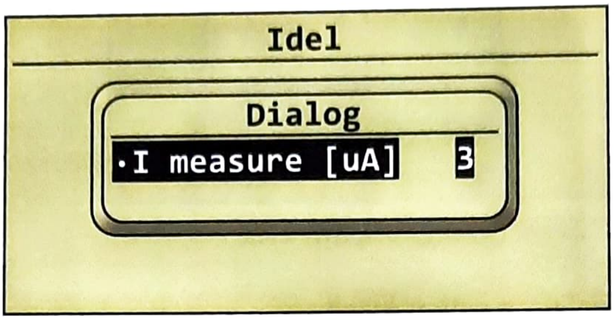


Fig .11. Current display

c. Enter the galvanostatic applied pulse current, **I measure [µA]**, and press Enter. The range is from 5 - 400 µA. Default is 100 µA.

3.5 Setting the reinforcement parameters.

a. Press the **Menu** key to enter the **Settings** menu. Fig. 12.

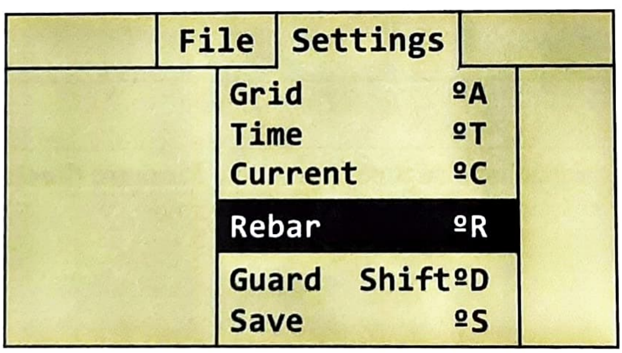


Fig .12. Settings display, rebar

b. Select **Rebar** and press **Enter**. Fig. 13.

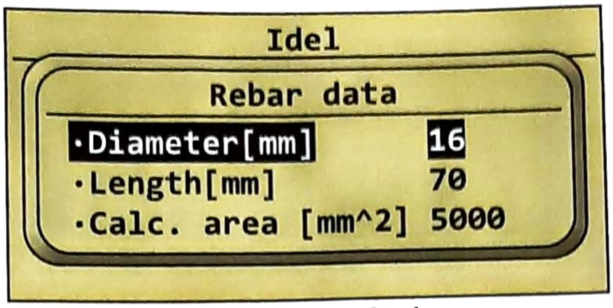


Fig .13. Rebar display

c. Use the arrow keys to select **Diameter [mm]** and enter the reinforcement diameter. The range is from 2 - 100 mm diameter. Default is 16 mm diameter.

d. Use the arrow keys to select Length []Using the standard electrode system and the guard ring, the **Length** is set by default to 70 mm, equal to the diameter of the electrode.

e. If the Guard Ring is not used, e.g. if the reinforcement is cut off at both ends, select **Length [mm]** and enter the actual length of the exposed reinforcement.

f. By moving the cursor down to **Calc. area [mm^2]** with the arrow keys, the correct area will be calculated. Alternatively, it is possible to enter another value for the **Calc.area [mm^2]** directly.

g. Press **Enter** when the parameters are set.

3.6 Turning On/Off the guard ring system

a. Press the **Menu** key to enter the Settings menu. Fig. 14.

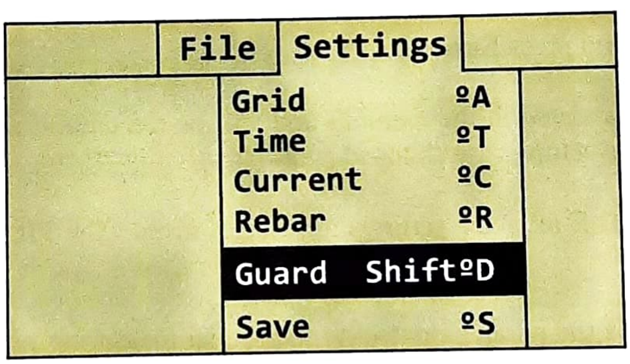


Fig .14. Settings display, guard

b. Select **Guard** and press **Enter**. Fig. 15.

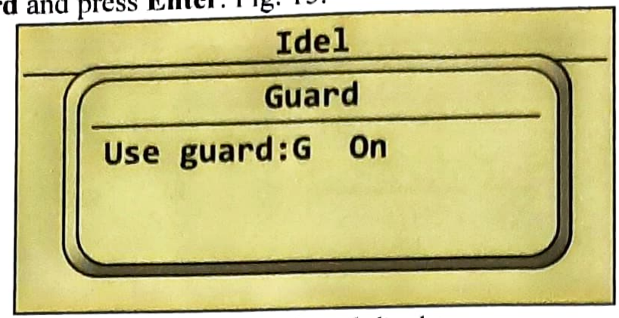


Fig .15. Guard display

c. Turn the Guard Ring on/off by pressing the **G** key at the keyboard. The Guard Ring should be set off only when an unconfined pulse is desired to make, e.g. if the reinforcement is cut off at both ends and the actual length of the exposed reinforcement has to be input.

d. Press **Enter**

3.7 Save the settings

a. Press the **Menu** key to enter the Settings menu. Fig. 16.

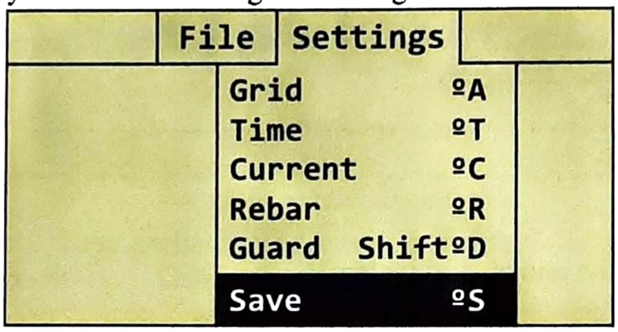


Fig .16. Settings display, save

b. Select **Save** and press **Enter**.

All information is now saved in the memory and will be the default parameters for all future measurements until the settings are changed again.

NOTE: Make your FILE after the settings have been saved. The FILE will then contain the settings.

It is possible to change the parameters for an individual measuring position during testing as described in section 3.12.

3.8 Calibrating the instrument

The instrument is calibrated at delivery, ready to be used, and should NOT be re-calibrated, unless:

- The FORCE program has to be re-installed and the software give the message that a calibration is required.

- The ambient temperature has been very extreme, affecting the stability of the electronics. Extreme temperatures are less than –10°C or more than +60°C.

NOTE: Should the calibration not be performed correctly, the measured values will be out of range.

If calibration is required the following setup of the equipment and procedure has to be applied:

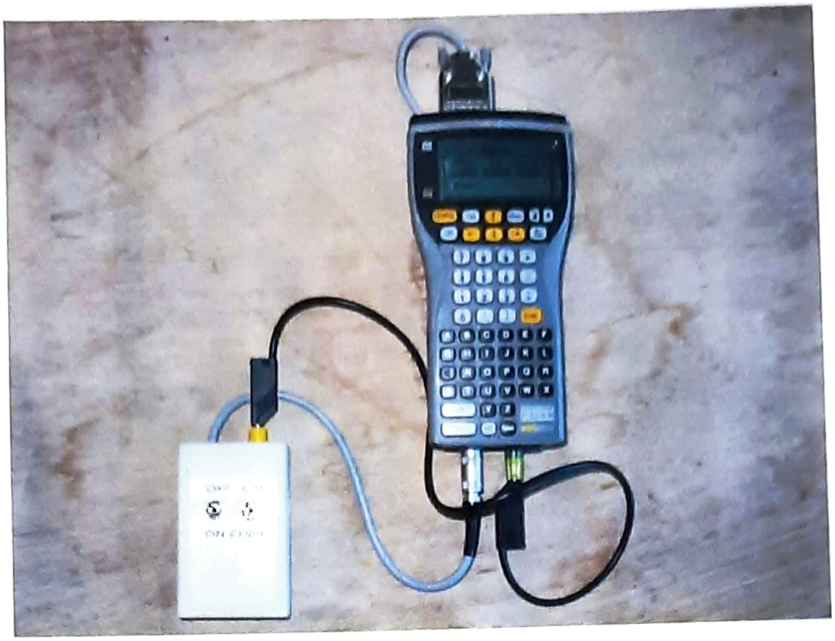


Fig 17. PSION and calibration box connected ready for calibration

The calibration procedure is as follows.

a. Connect the calibration box to the PSION. Fig. 17.
b. Check that the battery inside the calibration is fully charged, 9V.
c. Turn on the PSION. Select the **FORCE** program and press **Enter.**
d. Press the **Menu** key and select **Settings**.
e. Use the down ↓ key to select and press **Save.**
f. Use the down ↓ key again to select **Calibrate**. Fig. 18. (The **Calibrate** option is located below the **Save** option and will be visible when moving the cursor down through the menu).

5. Testing

When the electrical connection to the reinforcement is established, the PSION computer is connected to one clamping pliers by means of the black banana-banana cable and the cable drum cable. The electrode is also connected to the PSION computer.

Turn on the computer and follow the instructions from section 3 on how to set up the computer.

Fig. 33. The GalvaPulse Mark II cell ready for testing after wetting of the sponges

Fig. 34. The components of the Mark II cell (details in Appendix I)

The two sponges are wetted only with tap water and inserted in/on the electrode, fig 33 and 34. Details are given in Appendix I.

To make sure the electrode is working properly prior to testing, the check block can be used to verify that the readings are reasonable.

The sponges need to be wetted regularly during testing.

Lay out the grid on the testing surface. For accurate measurements, the grid points must be positioned directly above the reinforcement. Should measurement be needed where bars are crossing, the additional reinforcement area has to be taken into account.

If needed, spray water on the testing point as out lined in section 4.3.

Press the electrode firmly against the surface in the first testing point. The three attached screws in the electrode plastic disk have to rest against the surface and the electrode kept completely steady as long as the "clock" on the display of the PSION is running. Fig. 35.

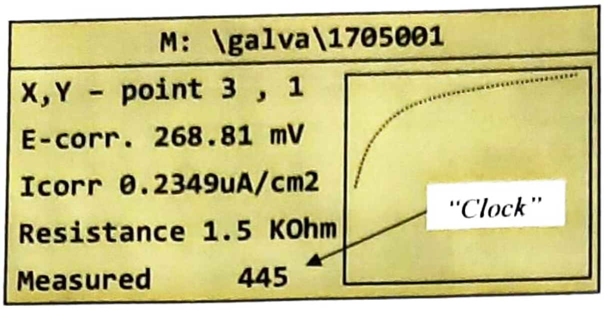


Fig. 35. If the "Measure time" display

When the "clock stops" the measurement has been taken and it is safe to remove the electrode to a new position.

The record is stored in the file created when the required arrow key used to move to a new testing point on the PSION keyboard is activated.

6. Practical steps on-site after testing

From the examination of the data additional testing is decided upon, e.g. for making repeated measurements in a smaller grid, chloride analysis, carbonation, impact-echo, strength testing or coring.

Where high apparent corrosion rates are found, the reinforcement is recommended to be visual inspected at a couple of locations to measure the actual reduction of the cross-section.

7. GalvaPulse Data Viewing and Reporting Program for Windows.

7.1 Installation

7.1.1 Windows 95/98/XP

a. Insert the CD-ROM in the CD-ROM drive of the office PC.

b. In the "Start menu" choose "Run" and type D:\ GalvInst.EXE ("D" being the CD-ROM drive of the PC, can be named by any other letter), and press Enter.

c. Follow the instruction on the screen until the installation is completed.

d. The program can now be launched from "Start" and "Programs" by clicking the "GalvaPulse" icon.

7.1.2 Windows 7/10

a. Insert the CD-ROM in the CD-ROM drive of the office PC and open Windows Explorer to navigate to the CD-Drive.

b. Right-Click on **GalvInst.EXE** and select **"Run as administrator".**

c. Follow the instruction on the screen until the installation is completed.

d. Before running the program, right-click on the shortcut from the Desk or on the shortcut from "Start" menu and click on "**Properties**". In the properties window, select "**Compatibility**" from the upper menu and check the box for the field "**Run this program in compatibility mode for:**" and select "**Windows XP (Service Pack 3)**" from the dropdown menu. Finally click the "**Apply**" button and then the "**OK**" button.

e. Launch the program either from the shortcut on the Desk or from the "Start" menu.

7.2 Using the program

a. Decide where to save the file and make a folder e.g. on the local hard disk. If no folder is made the files will automatically be stored in the directory where the program has been installed.

b. Connect the cable supplied to the PSION instrument's RS 232 terminal and to e.g. the COM1-port of the computer. If the computer doesn't have an RS 232 terminal, use the USB to Serial converter cable as extension. Download and install the dedicated driver if needed.

c. Launch the GalvaPulse Data Viewing and Reporting Program. The opening menu will be on the screen with the options: "**Get Data from Instrument**" or "**Get Data from File**". Select the relevant option. Fig. 36.

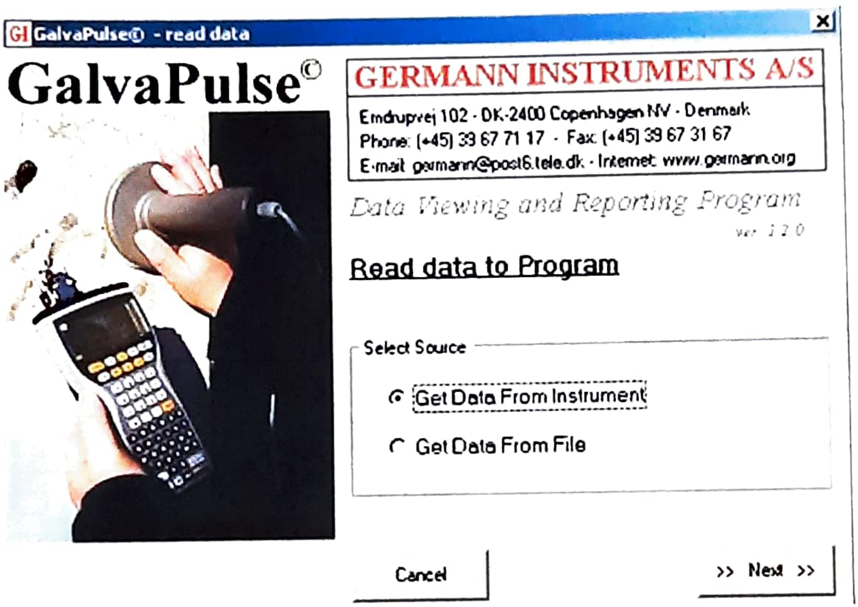


Fig. 36. Startup screen

7.3 "Get Data from Instrument":

a. Click the "**Next**" key. The "**Reading Data from Instrument**" screen will appear. Fig. 37:

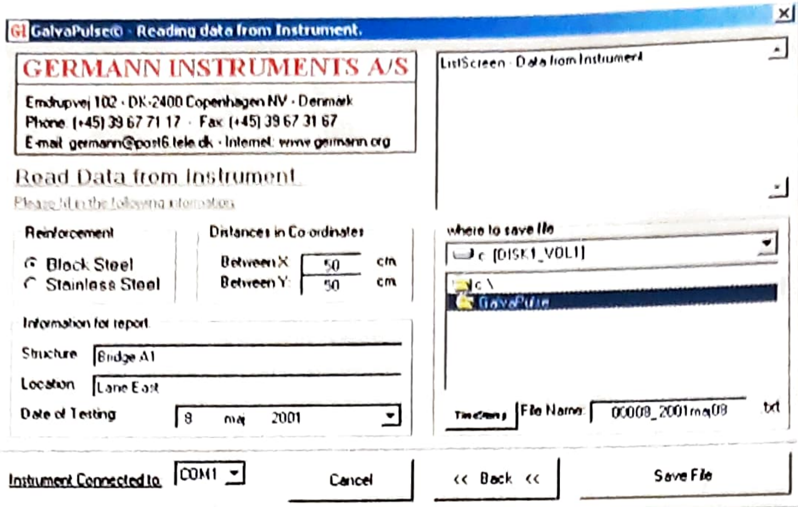


Fig. 37. Input screen

b. Chose the reinforcement type (Black Steel or Stainless Steel).

c. Enter the distance between the grid points.

d. Enter the information in the “Structure” and “Location” fields.

e. Select where to save the file (double click "**C:**", select the appropriate folder and double click on it with the mouse)

f. Type in the name of the file in the “**File Name”** box or press the “**Timestamp**” button. The last option will automatically write a number followed by the date.

g. Click with the mouse cursor in the center of the window called “**ListScreen – Data from instrument**”.

h. On the PSION computer, load the file which has to be transferred (see 3.10).

i. Once the file is loaded, press the “**Menu”** key of the PSION, go to "**Transmit**" and then "**Data Set**" by using the arrow keys of the PSION.

j. Press the “**Enter”** key on the PSION keyboard. The data in the file of the PSION will be transferred to the ListScreen. Fig. 38.

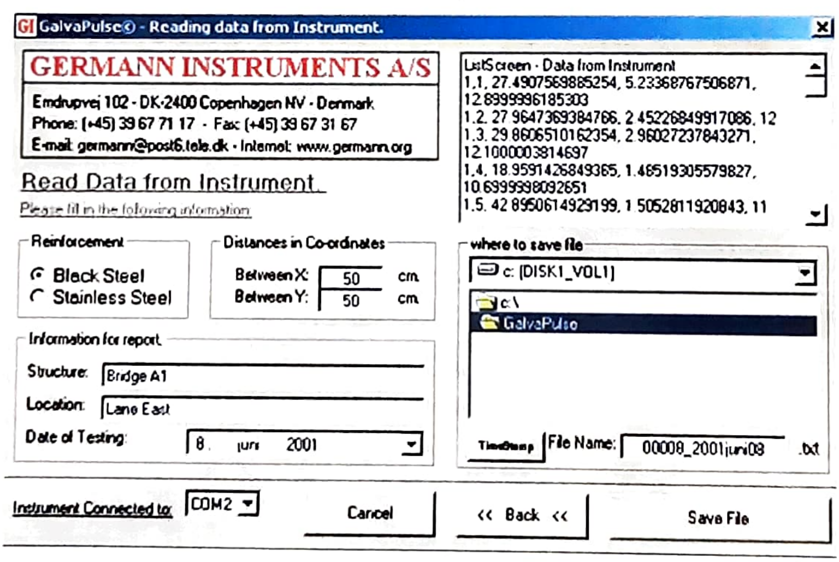
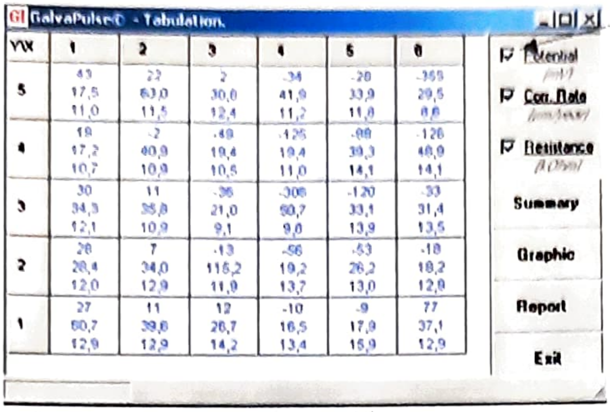


Fig. 38. List screen

k. When the transfer of data is finished, press the "**Save File**" button.

l. The data of the file is now presented in a table. Fig. 39.

By clicking on the √ data can be shown or hidden



Y\X	1	2	3	4	5	6
5	43 17,5 11,0	22 63,0 11,5	2 30,0 12,4	-34 41,9 11,2	-20 33,9 11,8	-355 29,5 8,6
4	19 17,2 10,7	-2 40,9 10,9	-49 19,4 10,5	-125 19,4 11,0	-88 39,3 14,1	-126 48,9 14,1
3	30 34,3 12,1	11 35,8 10,9	-36 21,0 9,1	-308 50,7 9,0	-120 33,1 13,9	-33 31,4 13,5
2	28 28,4 12,0	7 34,0 12,9	-13 115,2 11,9	-56 19,2 13,7	-53 26,2 13,0	-18 18,2 12,9
1	27 60,7 12,9	11 39,8 12,9	12 26,7 14,2	-10 16,5 13,4	-9 17,9 15,9	77 37,1 12,9

Fig. 39. Data presented in a table

7.4 Data presentation and report

GalvaPulse™ tabulation

The data collected are presented as numerical values, for the potentials in mV of Ag/AgCl electrode, the corrosion rate in µm/year and the electrical resistance in kOhm.

From the tabulation, a summary can be shown as well as the graphics (2D or 3D) and a report can be printed out.

Summary

The data are shown with the minimum, the average and the maximum values of the three sets of readings. Fig. 40:

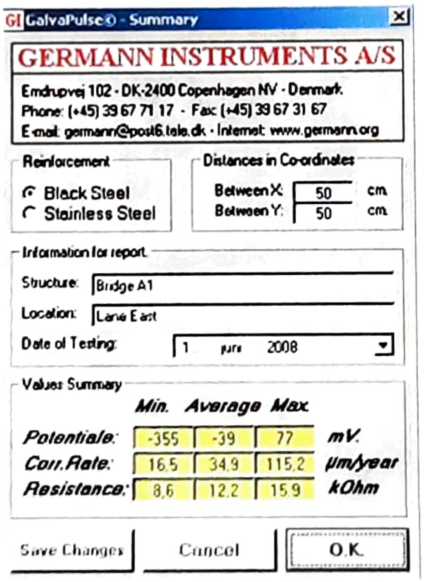


Fig. 39. Summary of data

Graphics
2D or a 3D plot can be selected for each of the three sets of readings. Fig. 41-42.

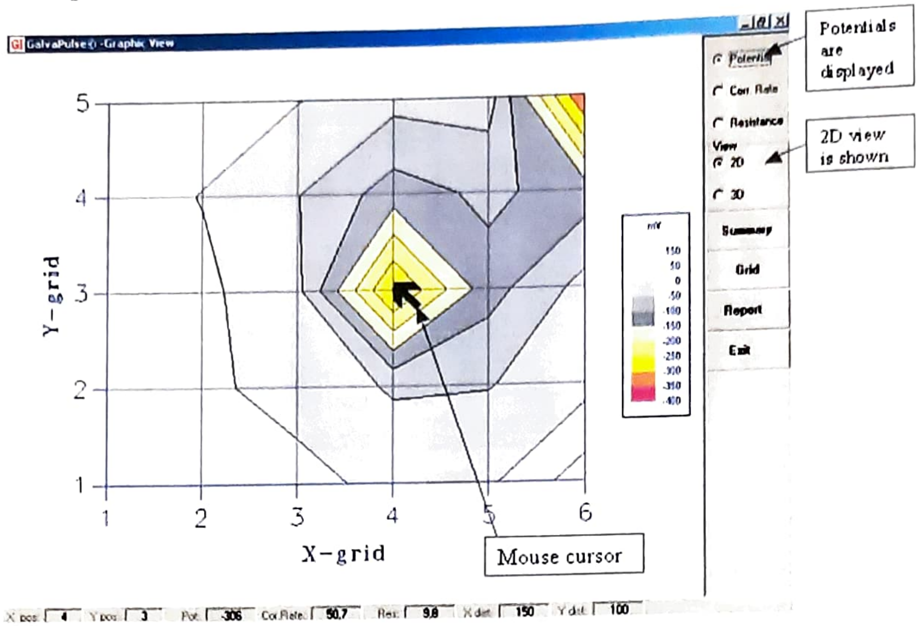


Fig. 41. Example of presentation of the mV-readings in 2D is shown above. The colours and limits are predefined for the corrosion rate and the potentials. For the resistance readings the instrument automatically sets the colours and limits. The predefined limits and colours can be altered upon request.

If the cursor is moved over the intersections of the grid, the data for each point is shown at the bottom of the screen. In this example the data shown are the X and Y position, Potentials, Corrosion Rate, Resistance and the distance from the staring point in the X and Y direction in cm (they are entered as mm in the equipment):

X pos.	*4*	*Y pos.*	*3*	*Pot*	*-306*	*Corr. Rate*	*50,7*	*Res.*	*9,8*	*X dist.*	*150*	*Y dist.*	*100*

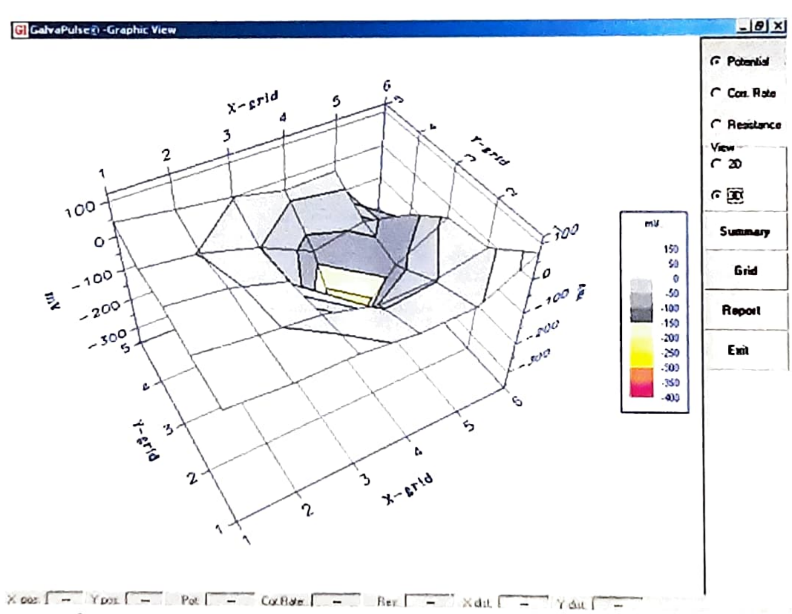


Fig. 42. Example of presentation of the mV-readings in 3D is shown above.

Report

The report options are: Front page with summary, grid containing the numerical data and the graphics.

7.5 "Get Data from File"

a. Click the "**Next**" button. The "**Open**" window will appear.

b. Choose the file in question, either by double clicking on it or typing in the name in the "**File Name**" window and clicking the "**Open**" button.

c. The data of the file is presented in a grid.

d. Data presentation and reporting are made as described earlier.

In appendix A the 9 pages illustrate a testing report.

8. Transfering raw data to a PC

The data collected is normally plotted in 2D or 3D graphs by using the supplied "GalvaPulse Data Viewing and Reporting Program" as described in section 7.

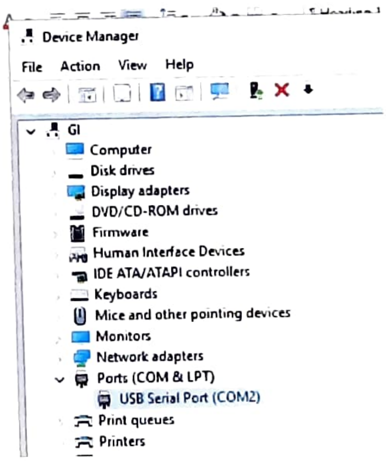
Alternatively, the raw data can be exported as text files, by downloading it with the Windows HyperTerminal. The HyperTerminal allows to download the raw data either of the individual polarization curves or of the testing results for further processing (e.g. in MS Excel) or for preventing loss of data in case the PSION computer is reset or runs out of the backup battery.

8.1 Transferring data of polarization curves or corrosion rates

a. To transfer data from the PSION to a standard PC, use the supplied cable and connect it to the RS 232 terminal of the PSION and a com-port on the computer. Use the USB-serial converter adaptor if the PC does not have a serial port.

b. If the program Psiwin2.1 is installed in your PC, make sure to deactivate it as it might block the com-port.

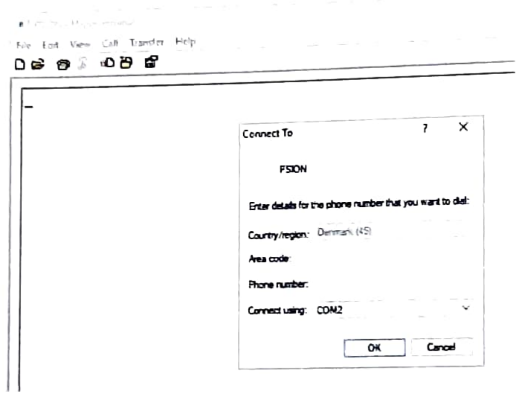
c. Connect the cable supplied to the PSION instrument's RS 232 terminal and to e.g. the COM1-port of the computer. If the computer doesn't have an RS 232 terminal, use the USB/Serial converter cable as extension. Download and install the dedicated driver if needed. Then go to the Device Manager of Windows to verify that the USB/Serial converter has been recognized and to check what COM-port number is assigned.



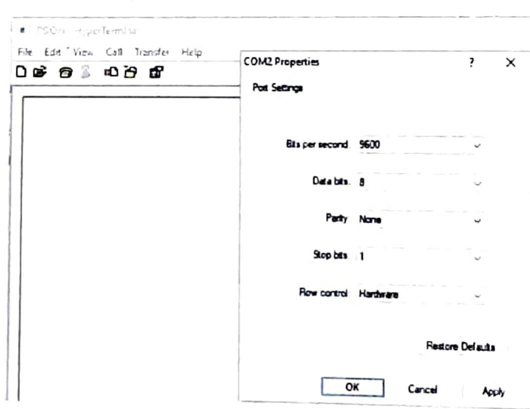
d. In the PC, go to Start/programs/Accessories/Communications/HyperTerminal and press enter. (If there is no HyperTerminal program installed in the PC it can be installed from the Windows installation CD-ROM). For **Windows 7 or 10**, the HyperTerminal has to be installed as it is not included in those versions of Windows. The HyperTerminal folder is included in the GalvaPulse CD delivered. Copy and save the folder in any desired location of the PC, open it and run the hypertrm.exe file (a shortcut can be created, e.g. on the Desktop).

e. Once the HyperTerminal is open, make a new connection and call it PSION.

f. Select the COM-port where the PSION is connected to and press **"OK"**.



g. As port settings, select 9600 Bits per second, 8 Data bits, No parity, 1 stop bit and Hardware flow control. Then press **"OK"** to continue.

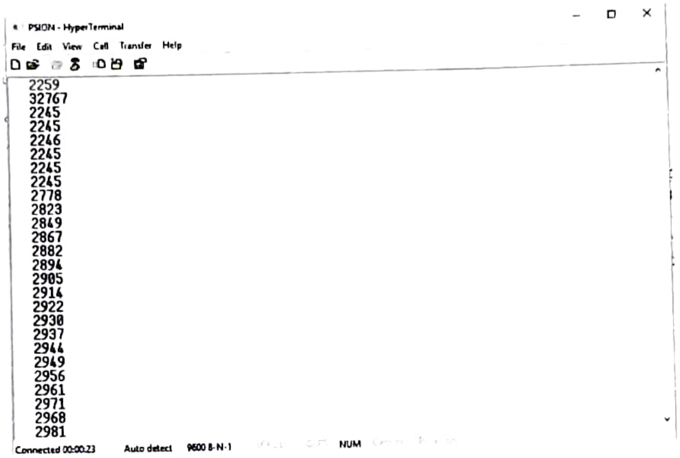


h. The PC is now ready to receive data from the PSION.

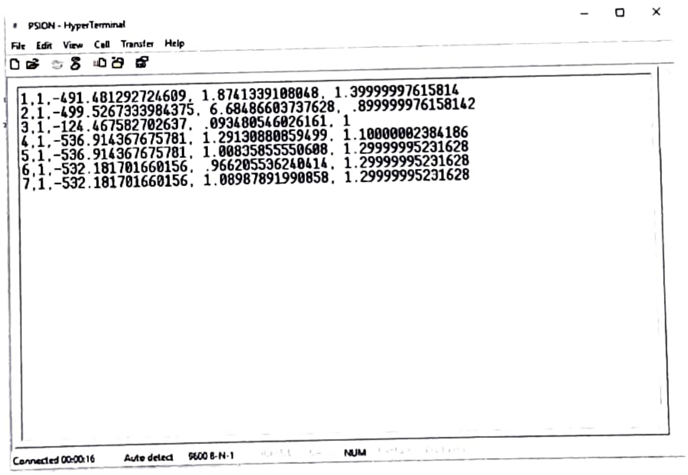
i. Load the data set to be transferred at the PSION (section 3.10) and press **"Menu"**

j. Select **"Transmit"** from the menu at the point X,Y of the pulse data set you want to send to the PC.

k. Select "**Measure data**" and press **"Enter"** to transmit the raw data of the polarization curve from the measurement at point X,Y to the PC. The first 11 numbers correspond to values of initial settings and end results, so the raw data to plot a polarization curve starts at row number 12. These data have to be converted to Volts by the conversion formula: $V = (15.5\ X / 65535) - K$, where X is the raw data value downloaded and K is a calibration constant whose value is usually 1.085, however, the value of K may be different from instrument to instrument and can change if a new calibration is made. Always obtain the value of K by solving the formula with the value of V equal to the value of E-Corr (mV) displayed from the test and the value of X equal to the first number of the data (the one at row number 12). The time lapse between two consecutive readings is 100 milliseconds. Copy and paste the data to the desired application (e.g. MS Excel).

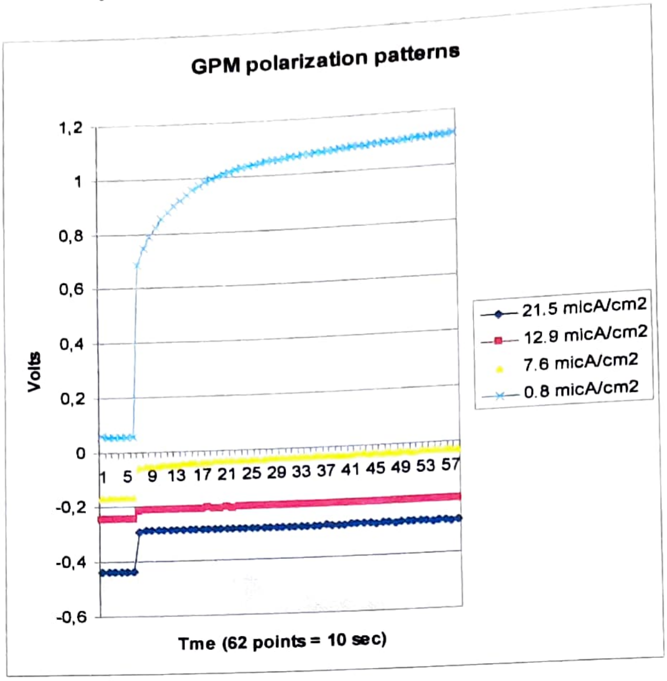


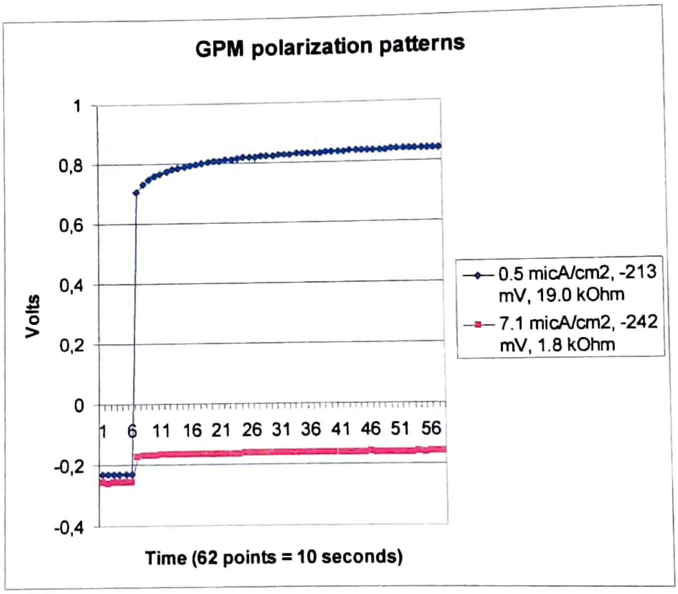
l. Select "**Data set**" and press **"Enter"** to transmit the testing results from the complete data set of X,Y points of the file. The values are presented in this order from left to right: E-corr (millivolts), I-corr ($\mu A/cm^2$) and Resistance (kOhms). Copy and paste the data to the desired application.



NOTE: The units for the corrosion rate (I-corr) raw data are $\mu A/cm^2$. The GalvaPulse software described in section 7 displays them in units of μm/year. Faraday's law of electrochemical equivalent can be used to find the conversion factor between these units: 1 µA/cm² corresponds to a penetration loss of carbon steel of approximately 11.6 µm/year.

The following are examples of polarization patterns from on-site testing that have been produced in MS Excel by transferring the raw data with the HyperTerminal application.





8.2 Transmitting data from HCP measurements

a. Follow the procedure in 8.1 up to point "i".

b. Load the data set of HCP measurements to be transferred at the PSION (section 3.10) and press "**Menu**"

c. Select "**Transmit data set**" at the HCP data set and press "**Enter**" and the Half Cell Potentials (millivolts) and Resistance (kOhms) from the complete data set of X,Y points of the file will be transmitted. Copy and paste the data to the desired application.

9. Maintenance

9.1 The PSION computer

The only maintenance of the PSION is to renew the two main batteries before a full day of testing is commenced. The Lithium back-up battery needs to be renewed approximately every 6-8 months.

Make sure either of the batteries (the set of two main batteries or the back-up battery) are functioning and in-place when the other is replaced, otherwise the FORCE program will be lost and has to be reinstalled, see Appendix B.

9.2 The electrode unit

Remove the sponges from the electrode unit immediately after every investigation is finished. Leaving the wet sponges in contact with the electrode's rings will corrode them and their lifetime will be shortened.

Clean both with tap water and squeeze them, gently, and let them dry. Place them in the suitcase adjacent to the electrode unit. Do not place them in a sealed plastic bag as this will only cause deterioration over time.

The self-contained reference electrode positioned inside the cell unit is long lasting and needs no special maintenance. For storing of the electrode after the testing, install the cap filled up with saturated KCl solution on the tip as outlined in Appendix I. In this manner the porous centre ceramic plug will not dry out and cause a bubble in the reference electrode.

The rings positioned in the circular plastic part of the electrode unit needs to be inspected before storage. If they have signs of corrosion on their surface, they need to be grinded to a smooth finish using the emery clothe. Flush afterwards with tap water and dry them with a piece of cloth. Do not use any other material.

9.3. Battery Replacement

The PSION computer contains two main batteries (1.5 Volt, size AA Alkaline) and a back-up battery (3 Volt, Lithium CR 1620 battery).

To replace the batteries, remove the plug with the cable from the pulse generator and then press the black knob on the backside of the PSION. The released tray containing the batteries will emerge. Pull it gently out.